

Multimodal Fashion & Context Retrieval

A decompose-and-verify hybrid for compositional fashion search · Glance ML Assignment

1,150-image corpus (Fashionpedia + Pexels top-up) · ~1.7 GB RAM and 120-190ms per query on CPU · index built on one GPU · live dashboard included

1. Problem and Dataset

The system retrieves images from a natural-language description of what someone wears, where they are, and the overall style. The hard part is compositional binding: "a red tie and a white shirt" must not match "a white tie and a red shirt." A single global CLIP-style embedding treats a caption as a bag of words, so it cannot tell the two apart.

The corpus is 1,000 Fashionpedia images, the dataset the assignment names directly, plus 150 Pexels photos that fill gaps Fashionpedia leaves in everyday settings. The top-up targeted exactly the graded axes: office scenes went from 0 to 12 images, home from about 2 to 30, park from 5 to 58, street from 15 to 155. Clothing types and colors were already broad in Fashionpedia. Even after the top-up, 895 of 1,150 images still carry no strong environment signal, since most Fashionpedia shots are runway or studio backgrounds; that gap is left for future work (Section 7) rather than masked with non-fashion filler images.

2. Approaches Considered

Approach	Binding	Cost	Best when
Vanilla CLIP or SigLIP, one embedding per image	Weak: cannot separate color from garment	Lowest	Loose semantic search where composition does not matter
Region-based: detect and crop each garment, embed each crop, match attributes	Strongest: binding is structural	High: needs a detector, 5-10x the vectors	Budget allows it and attribute precision matters most
Decompose-and-verify hybrid (chosen)	Strong: binding verified explicitly	Moderate: one vector per image plus a caption index	Fashion retrieval on a budget

The region approach sets the precision ceiling but pays for a detector and far more storage. Vanilla CLIP is free but fails the assignment's own compositional test case. The hybrid recovers most of the region approach's binding by verifying attributes against a cheap structured index instead of inferring them from one vector.

3. Chosen Architecture

```
QUERY "a red tie and a white shirt in a formal setting"
  parser.py splits it into clauses: [(red, tie), (white, shirt)], style=formal  (lexicon, no LLM, offline)
  FashionSigLIP encodes the full query -> FAISS top-50 candidates                (recall stage)
  FashionSigLIP encodes each clause -> minimum similarity across clauses        (binding penalty)
  attribute check compares query pairs against a SmolVLM-built structured index of {garments, environment, style}
  BM25 scores the same candidates against their captions
  final score = 0.35 dense + 0.25 min-clause + 0.25 attribute + 0.15 BM25 -> top-k results
```

FashionSigLIP is pretrained on fashion image-text pairs, so it resolves garment, fabric, and color vocabulary that a general CLIP model blurs together; it supplies recall and the ablation's strong baseline. **Query decomposition** runs on a deterministic lexicon with no LLM at query time, so it adds no

latency and stays reproducible. **The minimum across clause similarities** is the binding penalty: a candidate must score well on every clause, so an image with a matching tie but the wrong shirt is held back by its weakest clause. **Attribute verification** checks that a query's color and garment land on the same item in SmolVLM's structured caption record, which a single embedding cannot express. **BM25** catches exact terms the dense space can smear. Every signal is min-max normalized within the candidate set so the weights stay comparable.

4. Results

Ground truth is labeled automatically: an image counts as relevant when every parsed attribute of a query matches its structured caption. That same attribute check is also 25% of the hybrid scorer's weight, so the ablation below rewards the hybrid for agreeing with its own judge and should be read as a mechanism check, not an independent accuracy score. The hard-negative test that follows does not share that judge and is the stronger evidence.

Configuration	P@5	P@10	nDCG@10
Vanilla CLIP (ViT-B/32, dense only)	0.320	0.340	0.394
FashionSigLIP (dense only)	0.440	0.420	0.485
FashionSigLIP + clause min-pool	0.440	0.420	0.464
Full hybrid (+ attribute verification)	0.680	0.560	0.778

The dense-only comparison, FashionSigLIP against CLIP, is fair on its own since neither touches the attribute signal. One eval query has zero labeled ground truth: no image in this corpus captions both a red tie and a white shirt on the same person, a data-coverage gap rather than a modeling one. The hard-negative test below fills that gap with a corpus-independent measurement.

Scorer	Binding accuracy
Vanilla CLIP, single global embedding	0.450 (about chance)
FashionSigLIP dense	0.613
Full hybrid, explicit attribute verification	1.000

For 80 images that each contain two differently colored garments, the correct query ("a red dress and a blue jacket") is scored against its color-swapped twin ("a blue dress and a red jacket"), and the correct query must rank the image higher. Vanilla CLIP sits at chance because it sees the same words either way; the hybrid separates every pair correctly. This test still depends on caption accuracy, since it draws on the same SmolVLM captions the verifier checks against, but it does not depend on the ablation's judge, making it the cleanest evidence that explicit verification, not a lucky embedding, is what binds color to garment here.

5. Scaling to 1 Million Images

The flat FAISS index would move to an IVF-PQ index, cutting storage from 3 KB to 64 bytes per vector, about 64 MB total at 1 million images. Captions and BM25 would move from today's in-memory scorer, which scans every document per query, to SQLite FTS5, which scores only matching documents and keeps memory flat as the corpus grows. Dense and attribute rescoring already touch only the top-50 FAISS candidates regardless of corpus size, so that part of query cost stays constant; pre-filtering by environment and style before the vector search would shrink the candidate set further.

6. Zero-Shot Capability

Nothing here is trained on this corpus: FashionSigLIP retrieves from open-vocabulary text and SmolVLM captions images it has never seen. The parser lexicon generalizes to new combinations of known colors

and garments, so "a mustard trench coat in a park" binds correctly without that phrase ever appearing in training data. A color or garment entirely outside the lexicon falls back to dense-only scoring, which stays open-vocabulary through FashionSigLIP even when the exact-match verifier cannot fire.

7. Future Work

- **Locations and weather:** extend the SmolVLM schema with landmark and weather fields, reverse-geocode EXIF GPS at index time, and join a weather API on location and timestamp. Expose both as filterable metadata in a vector database such as Qdrant, so "someone in a raincoat in London while it's raining" pre-filters on city and weather before the dense search runs.
- **Precision, done:** the style field started at over 99% "other" since caption prose rarely says "formal" or "casual." Replacing that lexicon guess with a direct single-token classification lifted full-hybrid nDCG@10 from 0.400 to 0.778, the largest single lever found so far.
- **Precision, remaining:** fine-tune FashionSigLIP on Fashionpedia attribute pairs with swapped-color hard negatives, already mined by the eval pipeline, to target the binding failure directly; add a cross-encoder re-rank over the top 20 results; ground colors to garment regions with Fashionpedia's segmentation masks instead of caption text once budget allows a training run.

8. Codebase and Dashboard

Code: github.com/Neal006/multimodal-fashion-retrieval. The indexer and retriever live in separate directories, and `config.py` is the single source of paths, weights, and corpus selection, keeping logic separate from data. Running `python server.py` starts a FastAPI dashboard that loads the model and index once at startup and opens in the browser; each query only encodes the text and rescores the top-50 FAISS candidates, which keeps latency at 120-190ms on CPU with no GPU needed at query time, and the process holds about 1.7 GB of RAM once the model and index are loaded.